Pooled Mining

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Blockchain principles course (2025/03)

Preview

- High-level definition of the problem and solution
- How does a pool work?
- What is pool operator?
- What are the most common risks of giving too much power to pool operators?
- What are the most common ways to share the reward?

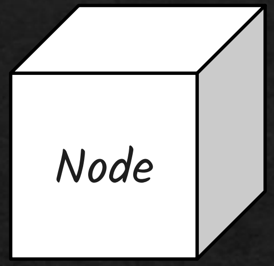
Problem

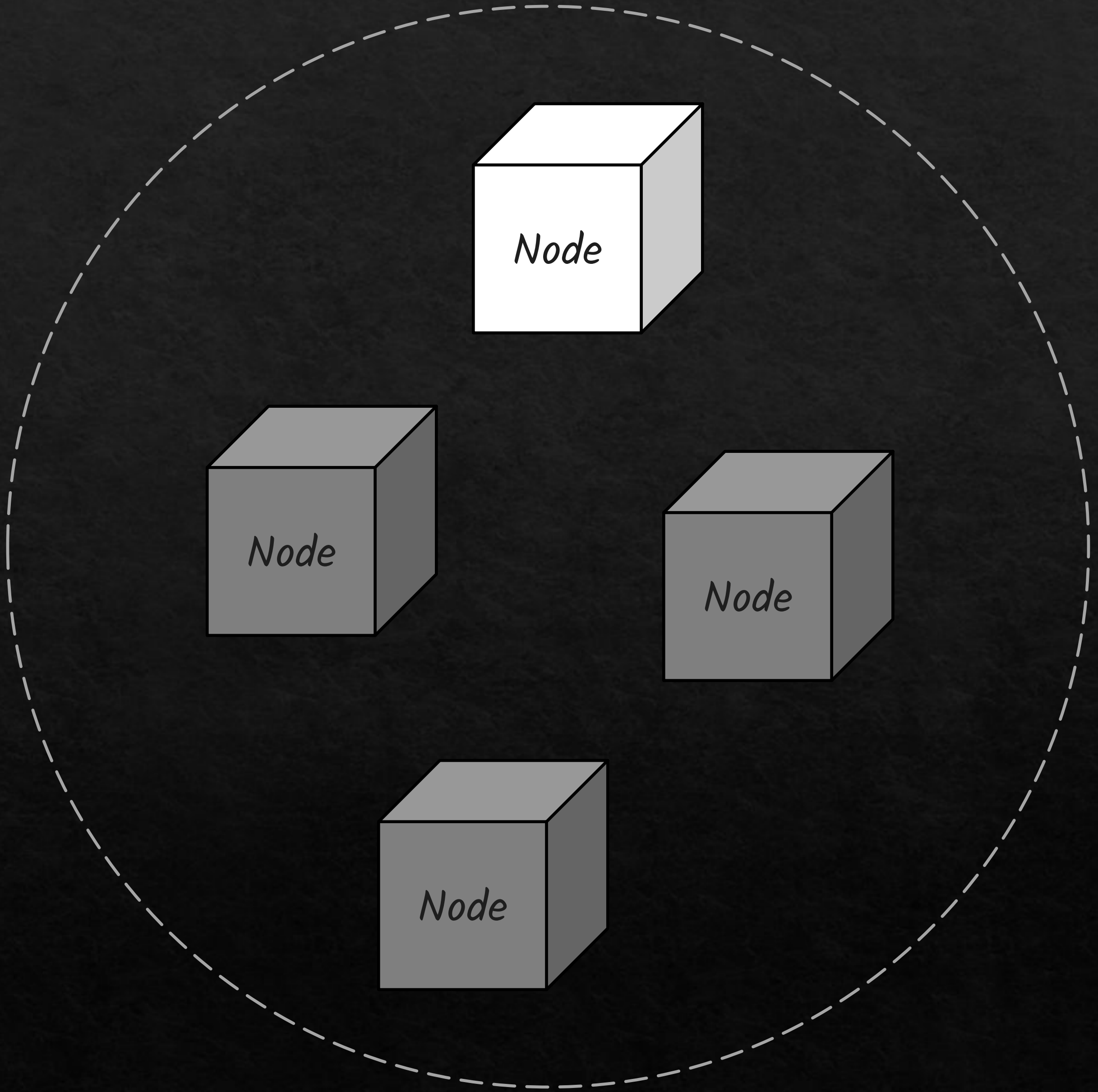
- Solo miners face **high variance** in rewards.
- Finding a block takes **millions of hashes**.
- Small miners can go **months without payout**.
- Result: discouraging for individuals and smaller setups.
- What is a trivial solution?

Current Difficulty	$D = 567358.224571$
Typical mining-rig speed	$R_{typ} = 500$ MHash/s
Probability of generating within fourteen days	$P_{single\text{-}hash} = 0.0000000000000004104346850315179415225053105586994206532$ $P_{single\text{-}second} = 1-(P'_{single\text{-}hash} \,\hat{}\, R_{typ})$ $P_{single\text{-}second} = 0.0000002052173214586726479267$ $P_{single\text{-}day} = 1-(P'_{single\text{-}hash} \,\hat{}\, (R_{typ} * 3600\ s/h * 24\ h/day)$ $P_{single\text{-}day} = 0.017574513073744846286208354 8$ $P_{fourteen\text{-}days} = 1-(P'_{single\text{-}day} \,\hat{}\, (R_{typ} * \ldots * 14\ day)$ $P_{fourteen\text{-}days} = 0.21982021908715036151288 15272$ $P_{fourteen\text{-}days} = $ **21.98%**
Average time to generate	56 days, 9 hours, 47 minutes

Solution

- Miners **combine hashpower**.
- Work on a **shared block template**.
- Receive **smaller, more regular payouts**.
- Pool operator manages work and reward distribution.


Node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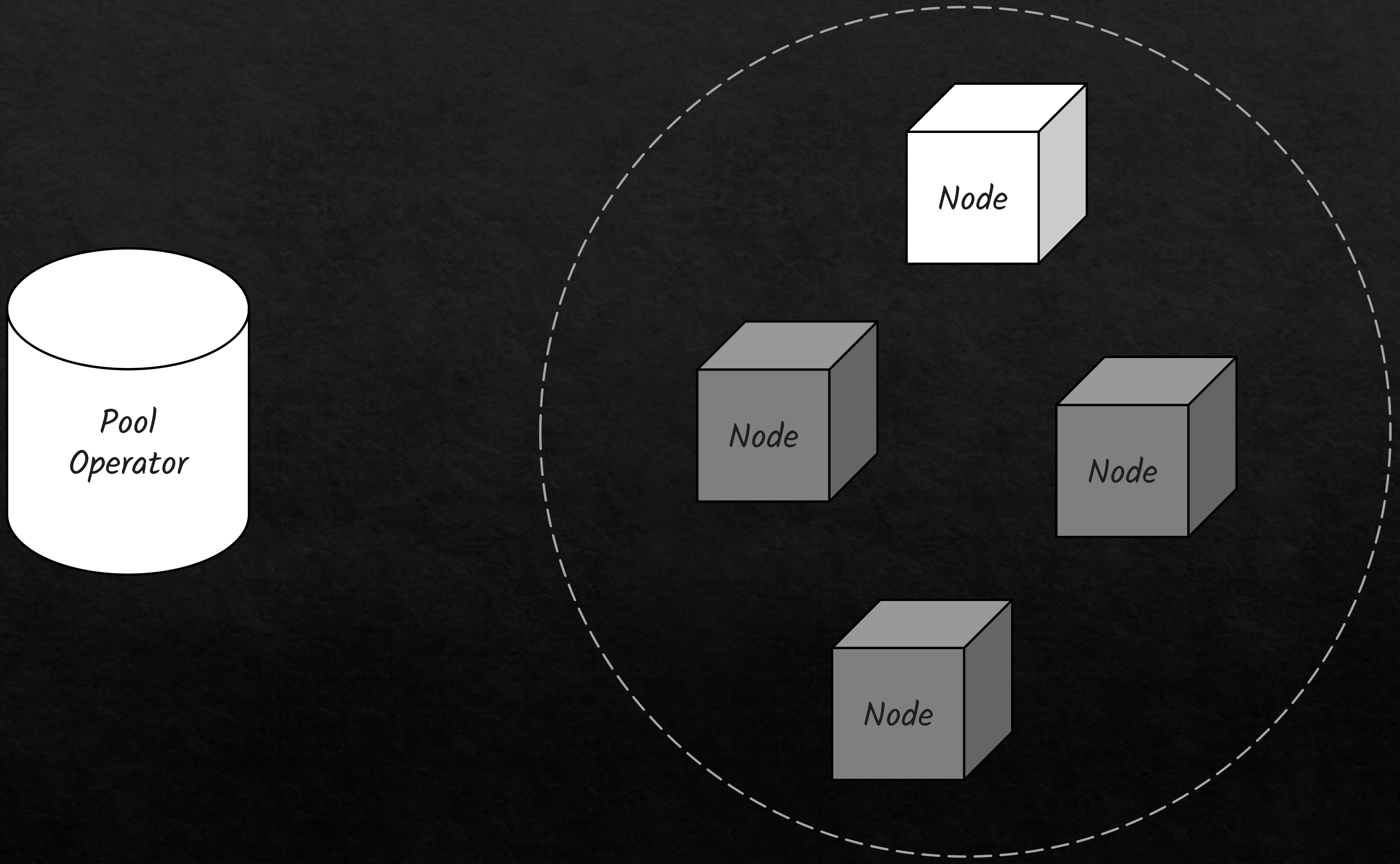
Pool
Operator
Node
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Node
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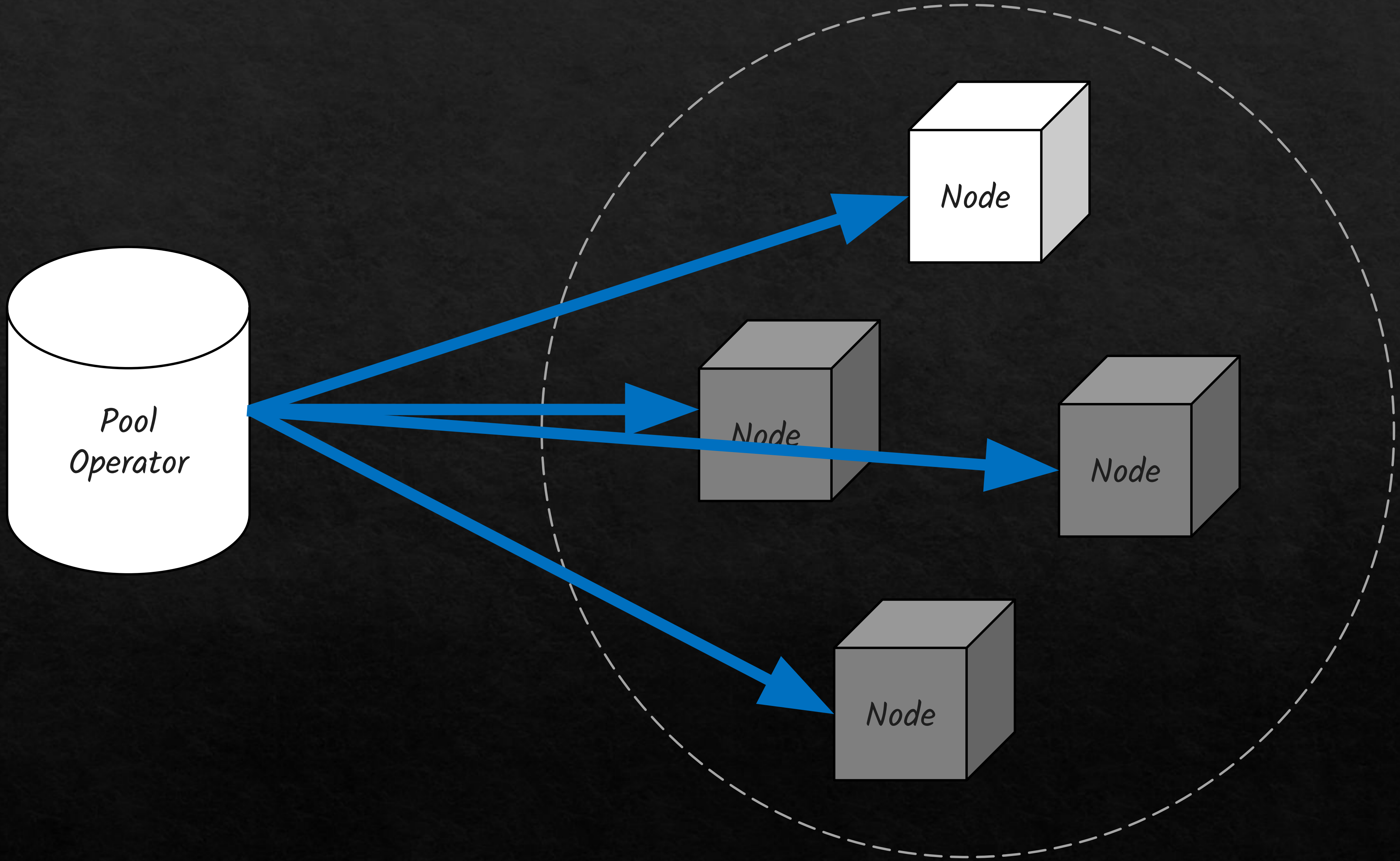
Pool
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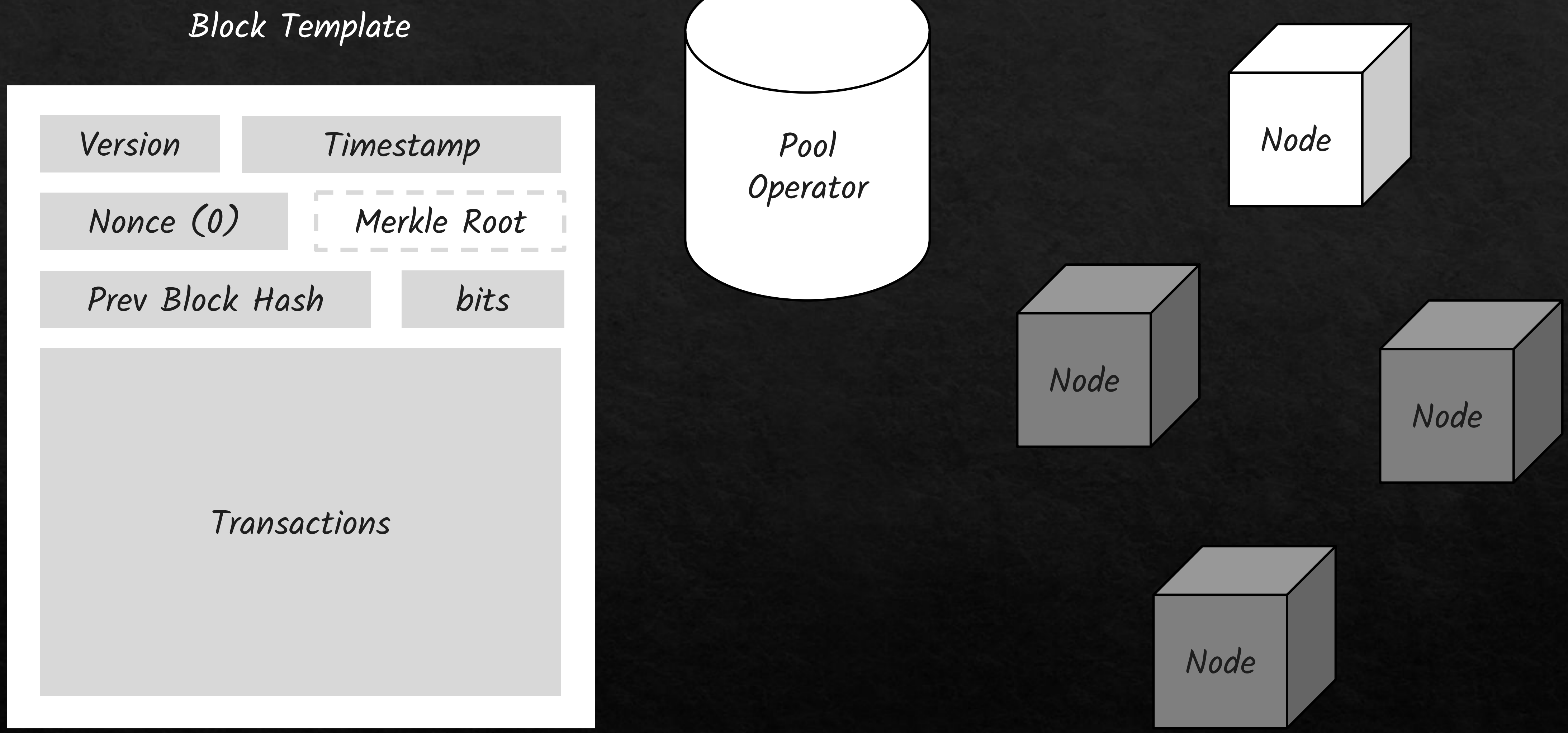
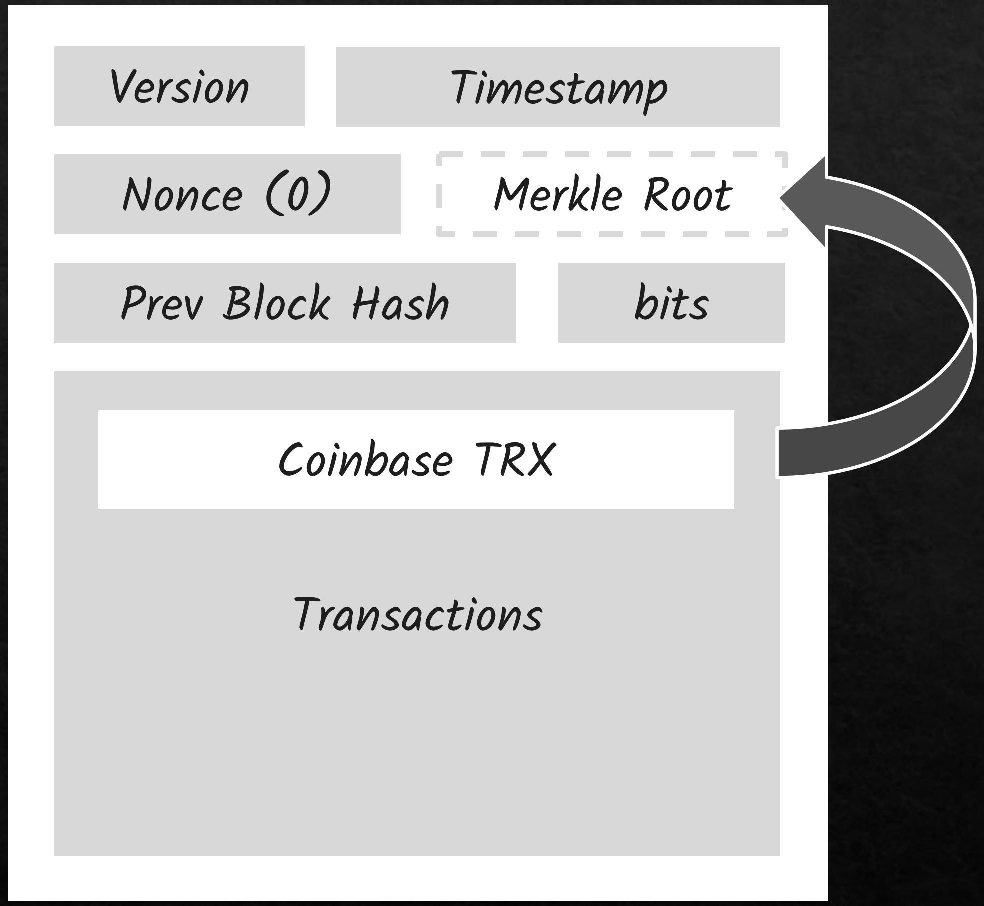

Block Template
Version
Timestamp
Nonce (0)
Merkle Root
Prev Block Hash
bits
Transactions
Pool
Operator
Node
Node
Node
Node

Block Template



Coinbase TRX Content

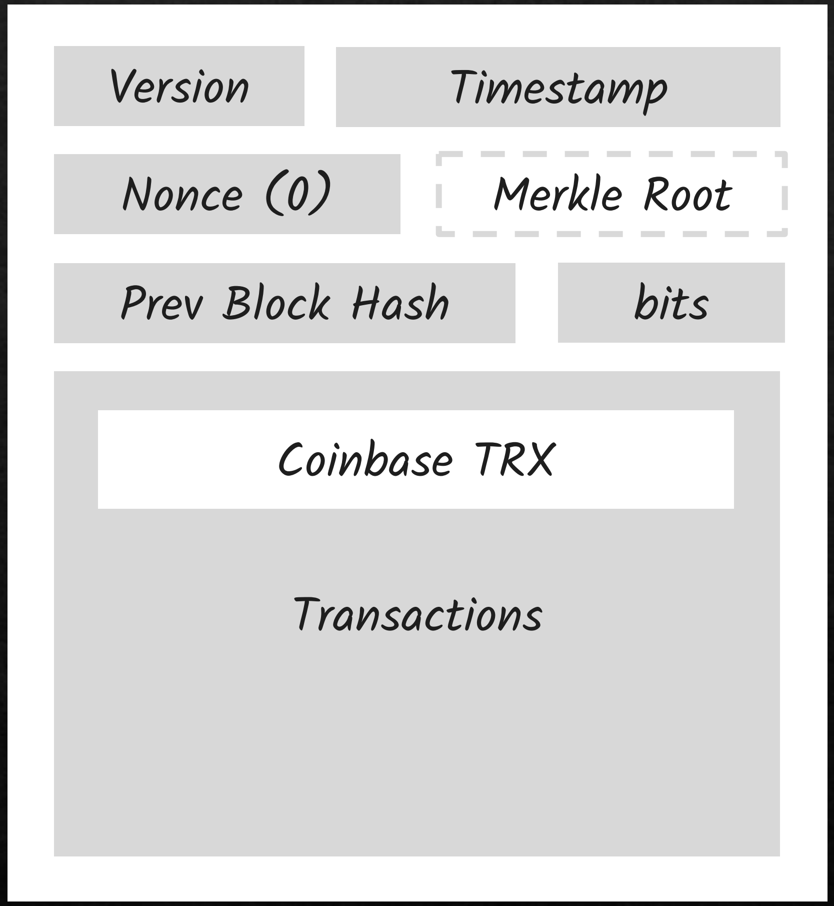
Output: Pool operator address

Amount: block reward + fees

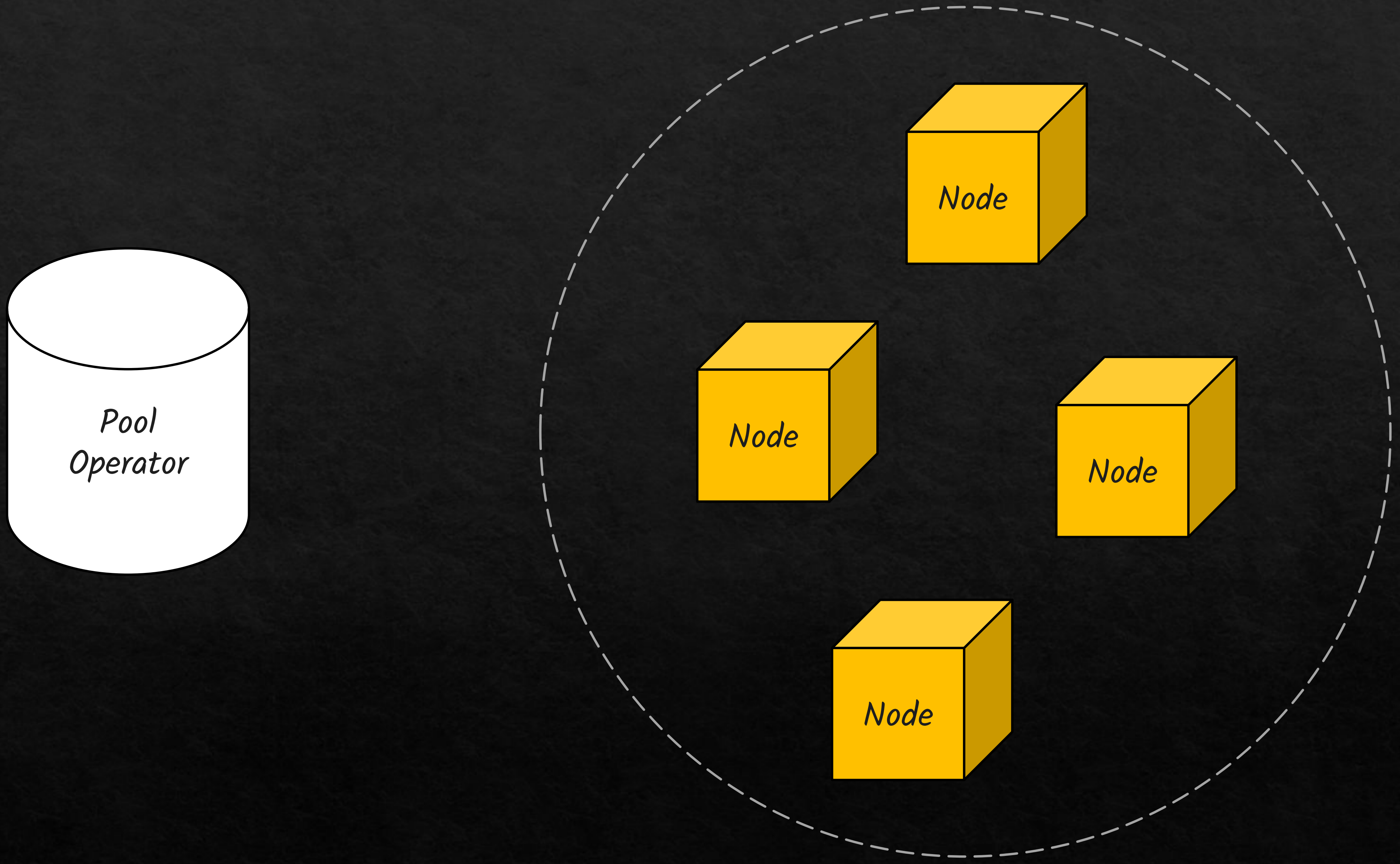
Pool **identifier or name**

Extra nonce space for miners to use

Block Template



The block template changes every few seconds as the operator receives new transactions or detects a new block on the network.
The operator then sends nodes the updated template to keep their work aligned with the latest state.


Pool Operator
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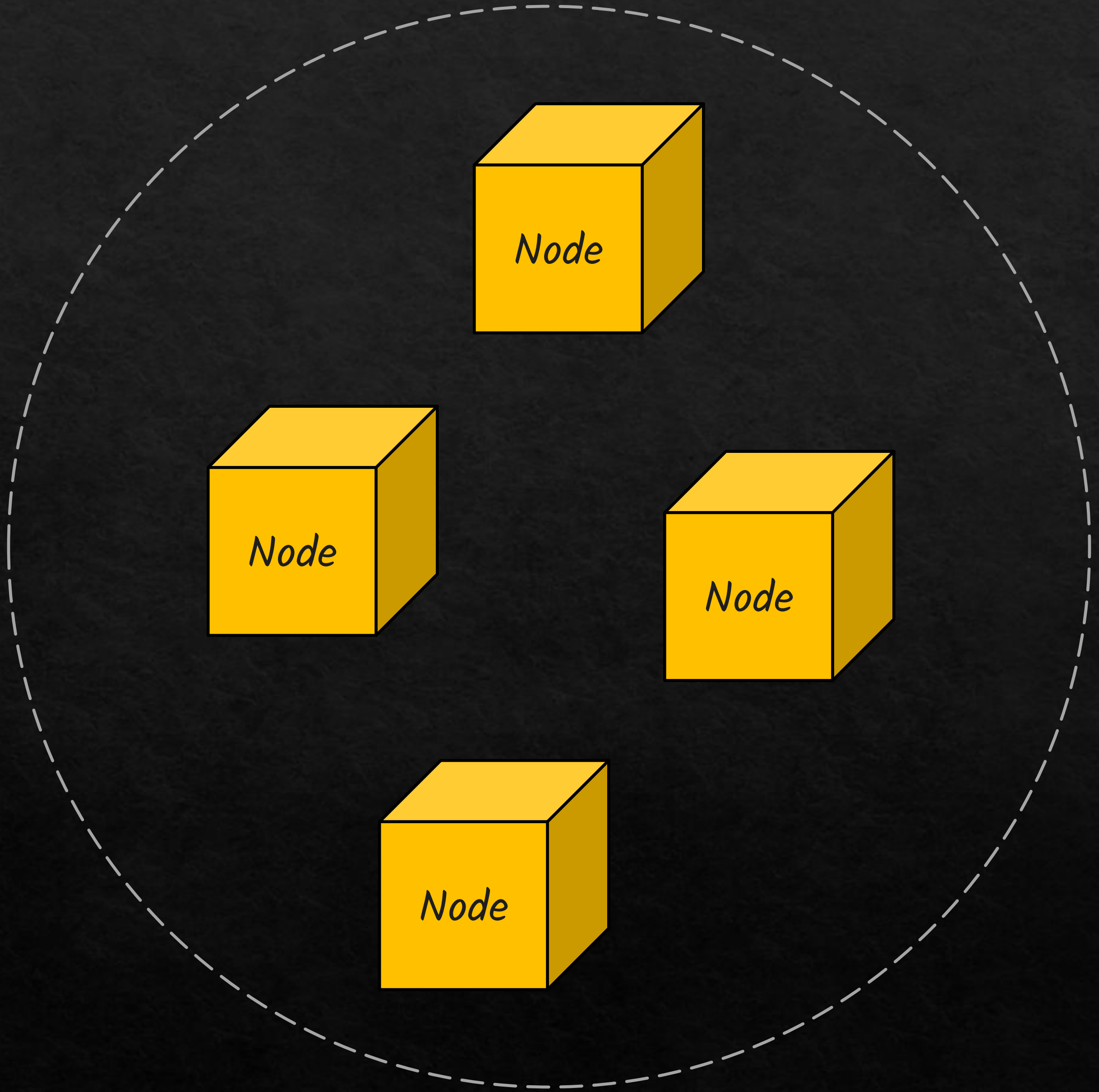
Node
Node

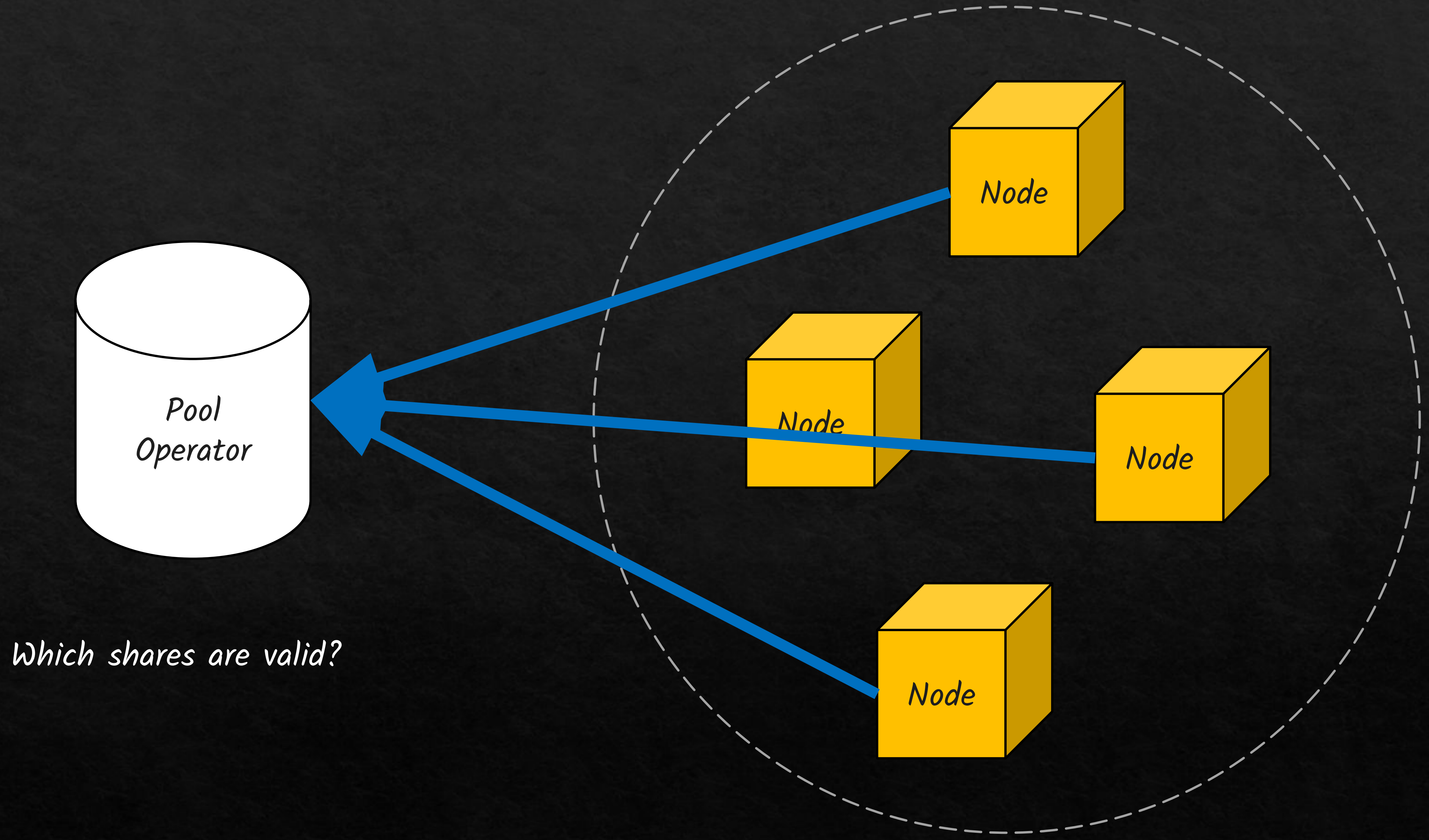


How to determine miners contribution?

There are several strategies to distribute the mining reward in a pool, but all are based on **proof of work**.

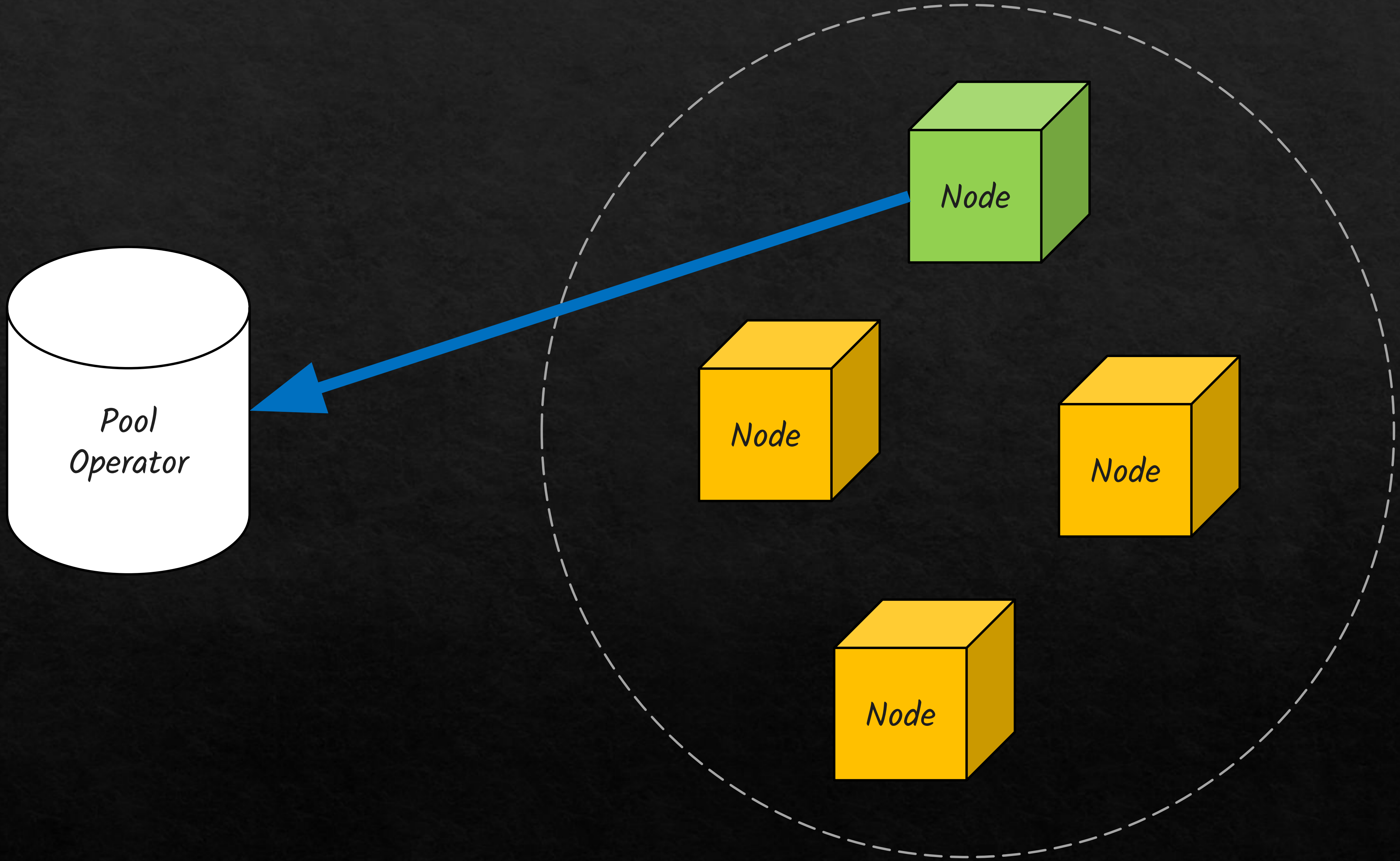
A common idea is to define a **lower difficulty threshold** for what the pool considers a "valid share", a hash that is easier to find than a real block but still proves work.

Each miner's reward is then proportional to the number of valid shares they submit over time.


Pool Operator
Node
Node
Node
Node
Which shares are valid?

Pool difficulty is communicated by the pool operator via another protocol

So miners know which results should be sent to the operator


Pool
Operator
Node
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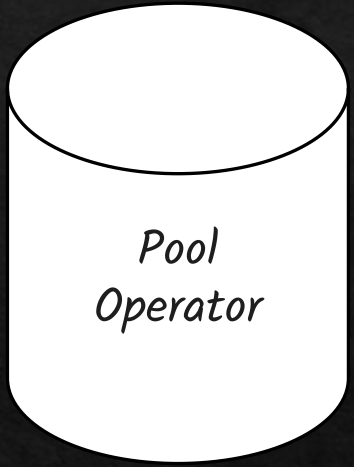
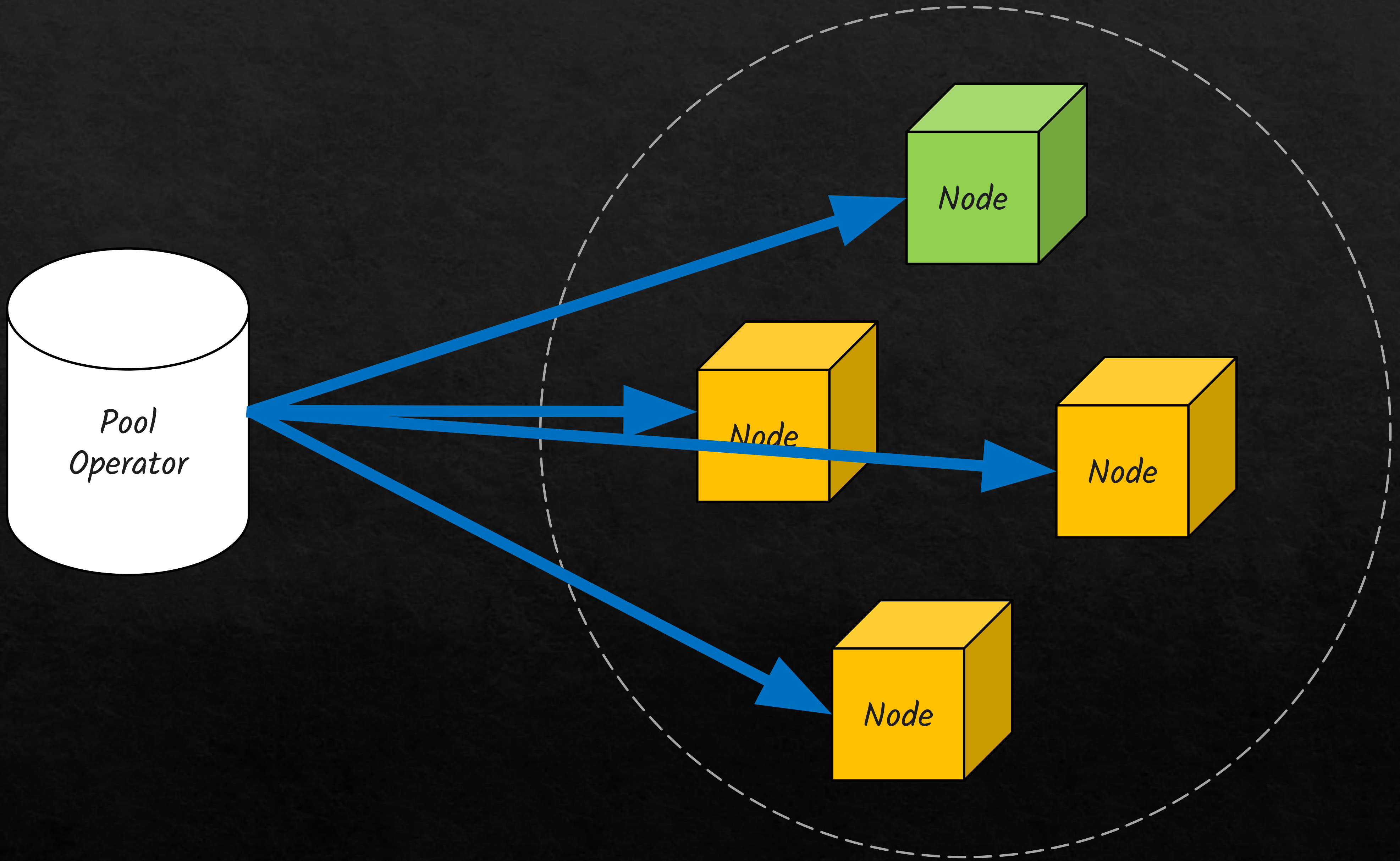


The pool operator then **broadcasts** the block with valid hash in the network and distributes the rewards according to its strategy, based on proof of work.



Pool
Operator
Node
Node
Node
Node

✅ Responsibilities of the Operator

Task	Description
1. Build block templates	Collect valid mempool transactions, construct candidate block templates.
2. Assign work to miners	Distribute templates and difficulty settings to miners.
3. Track miner shares	Validate incoming shares and attribute credit to the correct miner.
4. Broadcast valid blocks	If a miner finds a valid block, the operator immediately submits it to the network.
5. Distribute rewards	Once the block reward is confirmed, they pay miners based on the chosen reward scheme (e.g., PPS, PPLNS).
6. Enforce pool rules	Detect malicious miners (e.g., duplicate shares, selfish mining) and remove them.

⚠️ Can the Operator Disrupt the Network?

Censorship	The operator chooses which transactions go in the template. They could **exclude certain transactions** or addresses (e.g., blacklist behavior).
Empty Blocks	Some operators may prioritize block rewards and include fewer or no transactions to reduce propagation delays.
51% Risk	If a pool grows too large (over 50% hashrate), the operator could theoretically perform a **double-spend** or censor other miners.
Central Point of Failure	A compromised or malicious operator could **withhold valid blocks**, misreport shares, or vanish with funds.

How these risks are prevented? Reputation, Economic benefits and...

Proportional (PROP)

Proportional (PROP) rewards miners based on the number of shares they submitted during a single round.

How it Works:

A mining round begins when a new block template is issued.

Miners submit valid shares while the pool tries to find a real block.

Once the pool finds a block, the reward is distributed proportionally to each miner's share count in that round.

💰 Pros for Miners:

Simple and transparent.

Fair over time.

⚠️ Cons:

Income is inconsistent.

Vulnerable to pool-hopping, where miners jump in near the end of long rounds.

Pay-Per-Share approach (PPS)

Miners are paid a **fixed amount** for each valid share they submit, **regardless of whether the pool finds a block**.
How it Works: Operator defines a share difficulty (much lower than Bitcoin network difficulty).
Every submitted share proves work and earns a reward.
Reward per share ≈ (Block subsidy × Pool's success rate) / (Expected number of shares per block)

💰 Pros for Miners:
Predictable income, no variance.
Get paid instantly after submitting shares.
⚠️ Cons for Pool Operator:
Bears the risk of variance, must pay even if no block is found.
Needs a large buffer of coins to operate.

Full Pay-Per-Share (FPPS)

FPPS extends the classic PPS model by also distributing the **transaction fees** (not just the block subsidy) proportionally to miners.

How it Works: Pool calculates the expected block reward, including:
Block subsidy + Average transaction fees.

💰 Pros for Miners:
Stable and fair payout, including fees.
Immediate earnings without waiting for blocks.
⚠️ Cons for Operator:
Takes on full risk of variance in both block discovery and fee amounts.
Requires accurate fee estimation and financial reserves.

Pay-Per-Last-N-Shares (PPLNS)

PPLNS rewards miners based on the number of shares they submitted in the last N shares leading up to the pool actually finding a block.

How it Works:

Pool tracks the last N valid shares submitted before a block is found.

When a block is mined, the reward (block + fees) is distributed only among those who submitted shares in that window.

A miner's payout is proportional to their share count in that last N.

💰 Pros for Pool Operator:

Reduces payout risk.

Discourages "pool-hopping" (miners jumping between pools for short-term gain).

⚠️ Cons for Miners:

Unpredictable income.

Long-term miners benefit, but short-term participation might yield nothing.

Name	Location	Size[1]	Merged Mining[2]	Reward Type	Transaction fees	PPS Fee	Other Fee	Stratum
AntPool	China	Large	No	PPLNS & PPS	kept by pool	2.5%	0%	Yes
BTC.com		Medium	NMC	FPPS	shared	0%	4%	Yes
DEMAND	Global	Small	No	SOLO	shared		0%	Yes
F2Pool		Large	NMC, SYS, EMC	PPS+	shared	2.5%	0%	Yes
Golden Nonce Pool		Small	No	DGM	kept by pool		0%	Yes
KanoPool		Medium	No	PPLNSG	shared		0.9%	Yes
P2Pool	Global (p2p)	Small	Merged mining can be done on a "solo mining" basis [4]	PPLNS	shared		0%	Yes
Poolin	Global	Medium	NMC VCASH	FPPS	shared	2.5%	0%	Yes
SBICrypto Pool	Global	Medium	No	FPPS	shared	0%	0%	Yes
Slush Pool	Global	Medium	NMC	Score	shared		2%	Yes
Luxor		Medium	No	FPPS & PPS	shared	2%	0%	Yes
OCEAN		Small	No	TIDES	shared		0%	Yes

https://en.bitcoin.it/wiki/Pooled_mining

https://ieeexplore.ieee.org/abstract/document/8939805

https://btcmempool.org/graphs/mining/pools#6m

https://www.youtube.com/watch?v=6JytuGeMnKU

https://www.youtube.com/watch?v=J4AAXCaqlWc

Thanks